

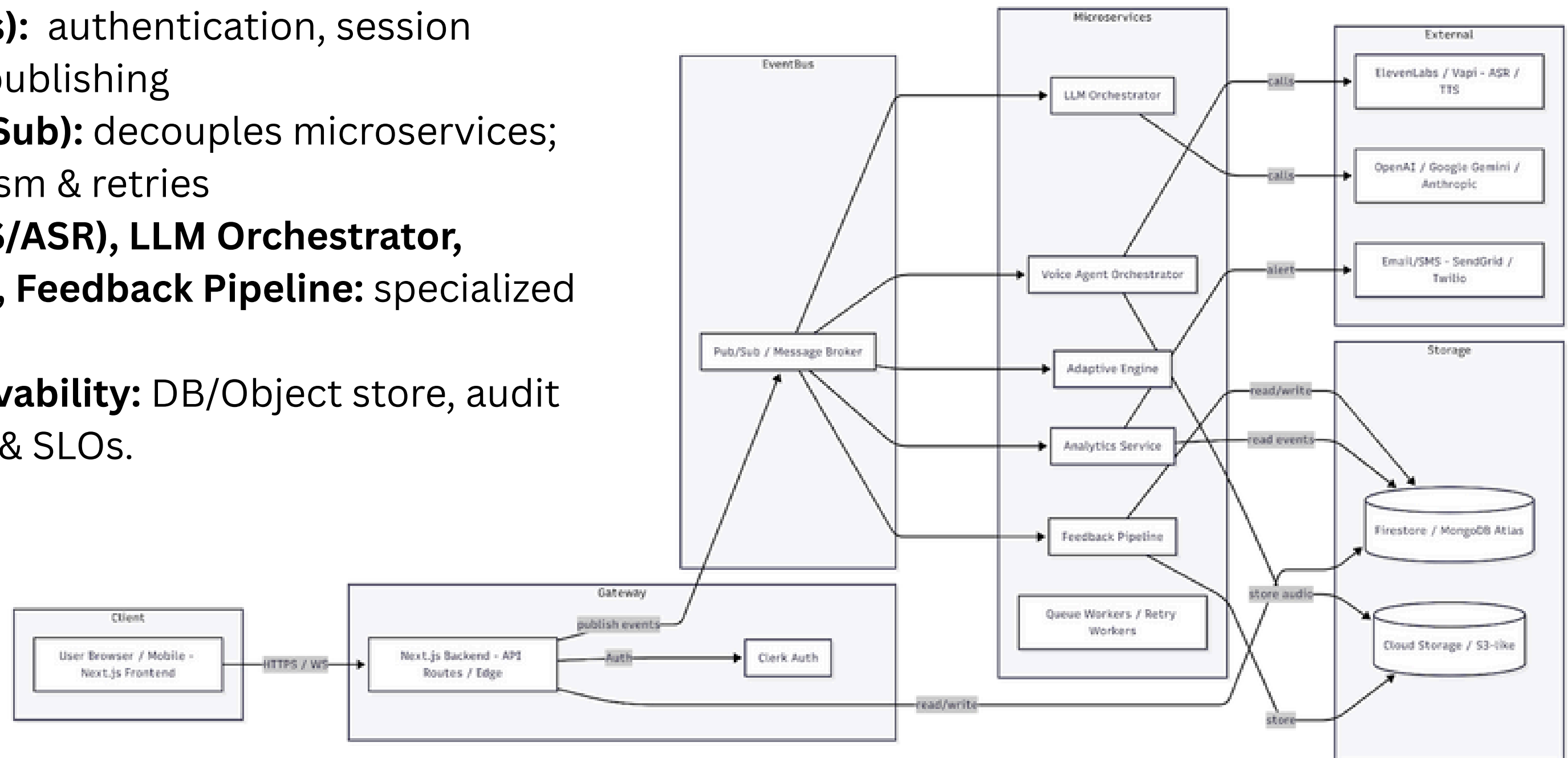
The background features decorative elements in the corners: top-left and bottom-right have overlapping teal and white geometric shapes with thin teal outlines; top-right and bottom-left have teal shapes with a grid of small teal circles.

# **AI INTERVIEW PREPARATION APP**

By P09

# SYSTEM ARCHITECTURE

- **Gateway (Next.js):** authentication, session lifecycle, event publishing
- **Event Bus (Pub/Sub):** decouples microservices; enables parallelism & retries
- **Voice Agent (TTS/ASR), LLM Orchestrator, Adaptive Engine, Feedback Pipeline:** specialized services.
- **Storage & Observability:** DB/Object store, audit logs, monitoring & SLOs.



# NON-FUNCTIONAL REQUIREMENTS

| Requirement                                      | Implementation in Architecture                                                                                                                                                                              |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low latency<br>( $\leq 2$ s follow-up)           | <ul style="list-style-type: none"><li>• Streaming ASR and pre-fetching of questions</li><li>• Event-driven microservices avoid blocking calls; Event-driven 'microservices</li></ul>                        |
| High availability<br>(99.5% uptime)              | <ul style="list-style-type: none"><li>• Managed multi-zone services (Vercel, Firebase)</li><li>• Circuit breakers and retries via event bus</li><li>• Dead-letter queues to recover failed event-</li></ul> |
| Concurrent scalability<br>( $\geq 500$ sessions) | <ul style="list-style-type: none"><li>• Stateless microservices scale horizontally</li><li>• Broker partitions distributing load</li><li>• Serverless autoscaling keeps costs predictable.</li></ul>        |

# SECURITY REQUIREMENTS

| Requirement            | Implementation in Architecture                                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Broken Authentication  | Clerk handles MFA, JWT, and role-based access. Next.js gateway validates all tokens. Private APIs are isolated in a secure network.             |
| Data Poisoning Attack  | Only verified user data is stored. Each report carries session IDs and timestamps. Feedback Pipeline validates provenance before training.      |
| Unsafe API Consumption | All API responses (ASR/LLM) validated using schemas. Outbound connections restricted to whitelisted domains. Content filtered for PII/toxicity. |



# WHY THIS WORKS

- Event-driven design = low-latency parallelism
- Serverless scaling = high concurrency
- Modular microservices = fault isolation
- Security layers integrated at every stage
- Supports both AI and human-in-the-loop feedback



The background features abstract geometric patterns in teal. In the top-left and bottom-left corners, there are nested rectangular outlines. In the top-right and bottom-right corners, there are clusters of small teal circles arranged in a grid. Diagonal teal lines cross the entire page, creating a dynamic, modern aesthetic.

# THANK YOU